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SECTION 01

FINAL PROJECT - What's Happening UTM?

GROUP 4 (DEBUGGERS)

GROUP MEMBERS	MATRIC NO
NUUR AISYAH BINTI RUZI	A23CS0168
NURUL NATALIA BINTI ROSNIZAM	A23CS0165
ADLINA NARISYA BINTI ISMAIL	A23CS0033
KOO XUAN	A23CS0300

LECTURER'S NAME : TS. DR. SARINA BINTI SULAIMAN

SUBMISSION DATE : 30 JUNE 2025

WEBSITE LINK: https://group4debuggers.infinityfreeapp.com

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1.0 Project Overview

Campus life at Universiti Teknologi Malaysia is rich in spontaneous food bazaars, club recruitment drives, cultural exhibitions, and academic forums. However, news of such events is fragmented, ad hoc, and too frequently discovered too late. “What’s Happening UTM?” answers that daily complaint by acting as a virtual megaphone for the campus community. The web application aggregates event promotion into one searchable database, from which students can discover what’s going on around them right now, and from which organizers can advertise to a broader, better-targeted audience without having to use dozens of separate ones. Immediacy is central to the concept, with anyone permitted to do so can post in minutes. Students seeking events such as late-night food offerings or midday lectures may browse by category and access full event details and contact information with a click. This frictionless sharing of information effectively eliminates the old low turnout caused by poster fatigue, chat-group overload, and word-of-mouth lag. System architecture hides complexity from users. Simple and user-friendly UI and lean JavaScript deliver responsive layouts and form validation that belong on any screen size. Behind the user interface, lean PHP scripts drive authentication, session management, and database transactions, maintaining data integrity and glitch-free CRUD operations. MySQL stores the foundation entities—users, posts, categories, and coordinates which supporting fast queries and relational integrity. The utilization of broadly taught technologies is designed such that future student developers can pick up and continue from the codebase with minimal onboarding. The project is primarily owned by the community. Rather of having administrative staff coordinate every event, it views students as both creators and consumers of knowledge, instilling a sense of shared responsibility. Campus size fosters peer-to-peer relationships and social interactions. Reading the feed is a daily discovery ritual, and broadcasting an event is an invitation rather than an obligation. Through considerate design, intelligent technological choices, and a clear focus for the user's benefit, "What's Happening UTM?" aims to transform unexpected meetings into purposeful interaction, enriching UTM's academic and social life one alert at a time.

2.0 Problem Statement

At Universiti Teknologi Malaysia (UTM), students often miss out on valuable campus events such as food sales, club activities, and academic talks due to the absence of a centralized communication platform. Event announcements are currently spread across various channels like WhatsApp, Instagram, Telegram, and physical posters, making it easy for information to be overlooked or forgotten. This fragmented communication leads to several issues, including low event turnout, reduced student engagement, and limited awareness of opportunities happening nearby. Organizers face challenges in promoting their events effectively, often having to repost the same content across multiple platforms without guaranteed reach or feedback. To address these challenges, there is a clear need for a centralized, real-time, and user-friendly system where students can easily discover and share campus events. By consolidating announcements into a single platform, the "What's Happening UTM?" website, aims to improve event visibility, increase participation, and enhance the overall campus experience.

3.0 Objective

The main objective of this project is to develop a centralized, community-driven web platform where UTM students can post, browse, and discover campus events in real time. By offering features like category filtering, location-based search, and responsive design, the system makes it easier for students to stay informed and engaged with campus activities. The platform also supports event organizers by allowing registered users to create and manage their own posts, upload images, and mark event locations on an interactive map. User authentication ensures secure access and that only the original poster can modify their content. Overall, this project aims to increase student participation, strengthen campus engagement, and serve as a practical application of full-stack web development skills in solving real-world problems.

4.0 System Architecture

The system uses a three-tier architecture:

- Frontend (Client-side): HTML, CSS (Bootstrap), and JavaScript for UI and interaction.
- Backend (Server-side): PHP scripts handle logic, CRUD operations, and session management.
- Database Layer: MySQL stores users, posts, categories, and locations.

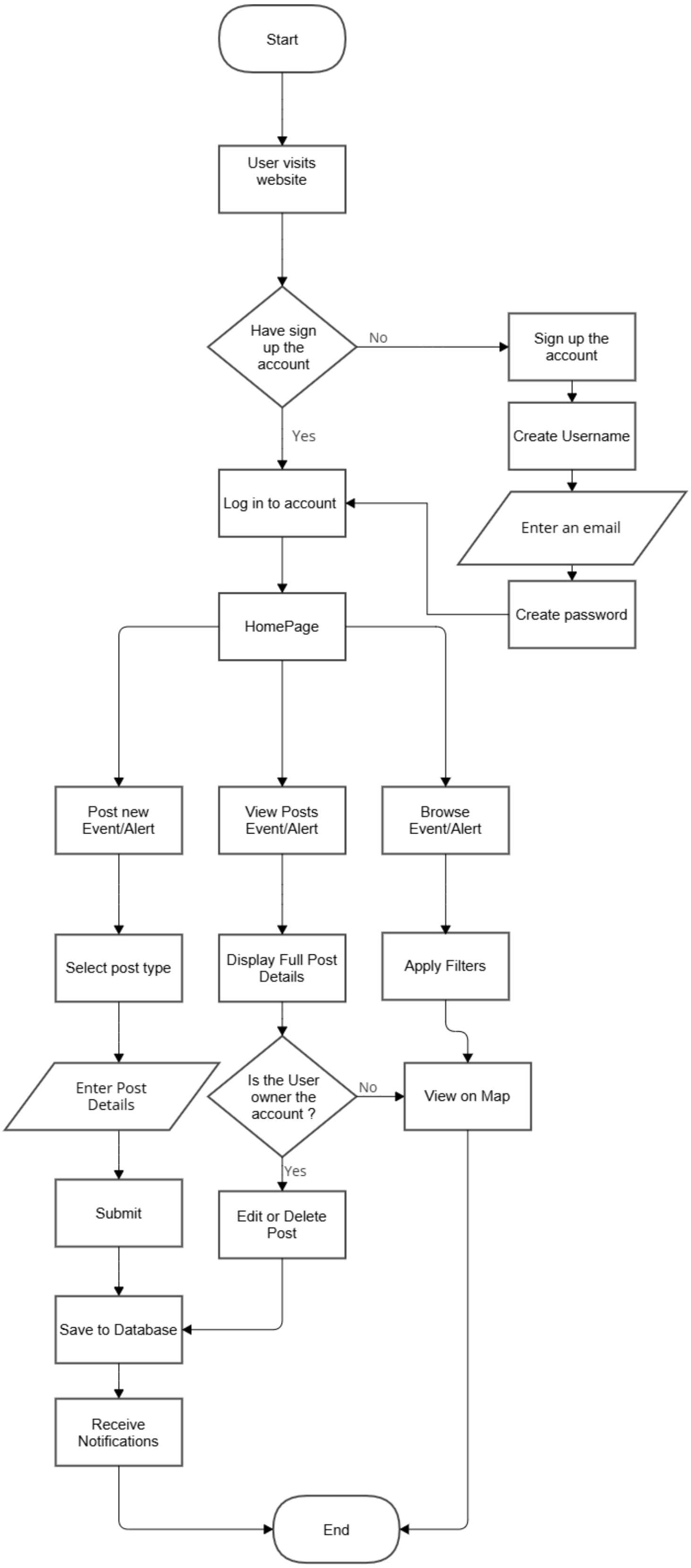
Interaction Flow:

- Users interact with the browser UI.
- Form submissions are sent via POST requests to PHP backend.
- PHP scripts perform CRUD operations with MySQL and return responses.
- Dynamic content is updated via PHP and JavaScript.

4.1 Technology Stack

Layer	**Technologies Used**
Front-End	HTML, CSS, JavaScript
Back-End	PHP (Core logic, session handling)
Database	MySQL (User and event data storage)
Optional API	Google Maps API (Map View), IP Geolocation

4.2 System Flowchart



5.0 Key Features

- **User Authentication**
  - Users can register for an account, log in securely, and access personalized features.
  - Features:
    ~ Registration and login forms with input validation
    ~ Passwords are securely handled
    ~ Sessions are managed using PHP to track logged-in users
  - Related Files:
    ~ register.html: Form where users input registration details
    ~ register_process.php: Validates and inserts user data into the database
    ~ login.html: Form for user login
    ~ login_process.php: Verifies user credentials and starts the session
    ~ logout_process.php: Ends the session and logs out the user
    ~ profile.html: Displays user profile after login
    ~ profile_data.php: Fetches user data from the database for display
    ~ config.php: Centralized database connection file

- **Event Management (CRUD)**
  - Users can manage event posts by creating, reading, updating, and deleting them.
  - Features:
    ~ Users can submit a post including title, description, date/time, category, and location
    ~ Edit and delete functions are available only for the post owner
    ~ Posts are stored in a MySQL database and rendered dynamically
  - Related Files:
    ~ post-event.php: Enable users to post new events
    ~ edit-event.php: Enable users to edit or update events details
    ~ delete-event.php: Enable users to delete events that unable

- **Sorting and Filtering**
  - Users can easily find relevant events using sorting and filtering tools.

- Features:

  ~ Filter events based on selected category (e.g., food fair, club event, academic talk)
- Related Files:

  ~ event_view.html: Displays event list and includes dropdowns or filter bars.

  ~ index.php: Handles user filter input (e.g., category) and queries the database for matching events.

● **Responsive Design**
- The system adapts fluidly to any screen size: desktop, tablet, or mobile.
- Features:

  ~ Uses CSS media queries

  ~ Ensures login form, event list, and navbar are mobile-friendly
- Related Files:

  ~ style.css: Contains the custom styling, including layout and responsive adjustments

  ~ some include the css style inside the file like login.html, index.php, and other UI pages utilize the styles

● **Validation and Error Handling**
- Ensures only correct, safe data is submitted through forms.
- Features:

  ~ Client-side validation: Checks for empty fields, invalid input using JavaScript before submission

  ~ Server-side validation: Re-validates data and handles errors using PHP (e.g., existing username, login failure)

  ~ Feedback messages: Clear messages displayed if login fails, fields are missing, etc.
- Related Files:

  ~ auth.js: Handles JavaScript validation logic

  ~ login_process.php, logout_process.php, register_process.php, config.php, profile_data.php, post-event.php: Perform PHP validation

  ~ HTML pages use <div id="message"> blocks for feedback

- **User-Friendly Interface**
    - Focus on intuitive navigation and clarity in design.
    - Features:

      ~ Clean layout with consistent styling

      ~ Users receive feedback on actions like “Event Created” or “Login Failed”

      ~ Simple navigation bar and footer for all users
    - Related Files:

      ~ style.css: Defines layout, color schemes, fonts, button styles

      ~ All HTML pages (login, profile, index) reflect the styling

- **Map View**
    - Display event pins on Google Maps
    - Features:

      ~ Interactive map with pins for each event's location

      ~ Events may include coordinates or addresses linked to Google Maps
    - Related Files:

      ~ map.html: Enables map functionality and shows events as interactive pins.

- **Photo Upload**
    - Users can attach images to their posts for better visibility.
    - Features:

      ~ Upload event banners or promotional posters

      ~ Images stored on the server and displayed with posts
    - Related Files:

      ~ post_event.php: Allows users to upload images during event creation.

      ~ event-view.html: Retrieves the image path and shows it next to the event post.

6.0 Database Design

Database Name: **utm_events**

Tables:

1. **users**

```
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(50) NOT NULL UNIQUE,
  email VARCHAR(100) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

2. **events**

```
CREATE TABLE events (
  id INT AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(255) NOT NULL,
  location VARCHAR(255) NOT NULL,
  datetime DATETIME NOT NULL,
  description TEXT NOT NULL,
  poster VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

7.0 Setup and Deployment Instructions

This section provides detailed instructions for setting up and deploying the "What's Happening UTM?" web application both locally (for development and testing) and online using the InfinityFree free hosting service.

7.1 Local Testing

For development and testing purposes, the application can be run locally using XAMPP.

Step 1: Install XAMPP

Install XAMPP on local machine

Step 2: Run Locally

1. Copy the project folder into the appropriate web directory which is into htdocs for XAMPP
2. Start the Apache and MySQL servers using the control panel (XAMPP)

Step 3: Import the SQL database

Import the SQL database which is utm_events.sql into phpMyAdmin

Step 4: Configure database

Configure database connection by editing the config.php file:

```
<?php

$host = "localhost";

$user = "root";

$pass = "";

$dbname = "utm_events";

?>
```

Step 5: Running Application

Run the application in browser

7.2 Online Deployment Using InfinityFree

The project is hosted online using InfinityFree, a free hosting provider. The steps below explain how the deployment was completed.

Step 1:Creating a hosting account

Create a new hosting account and register a free subdomain

Step 2: Uploading Project Files

1. Open File Manager or connect via FTP client

2. Upload the entire project folder to the htdocs directory

Step 3: Setting Up the Online Database

1. In the cPanel, go to MySQL Databases
2. Create a new database
3. Click phpMyAdmin, select the new database, and import the utm_events.sql file

Step 4: Configuring the Online Database Connection

Edit the config.php file with the actual infinityFree database credentials:

```
<?php

$host = "sqlXXX.epizy.com";     // Replace with the actual hostname from InfinityFree

$user = "epiz_12345678";        // Your InfinityFree database username

$pass = "your_password";        // The password you set during DB creation

$dbname = "epiz_12345678_utm_events";

?>
```

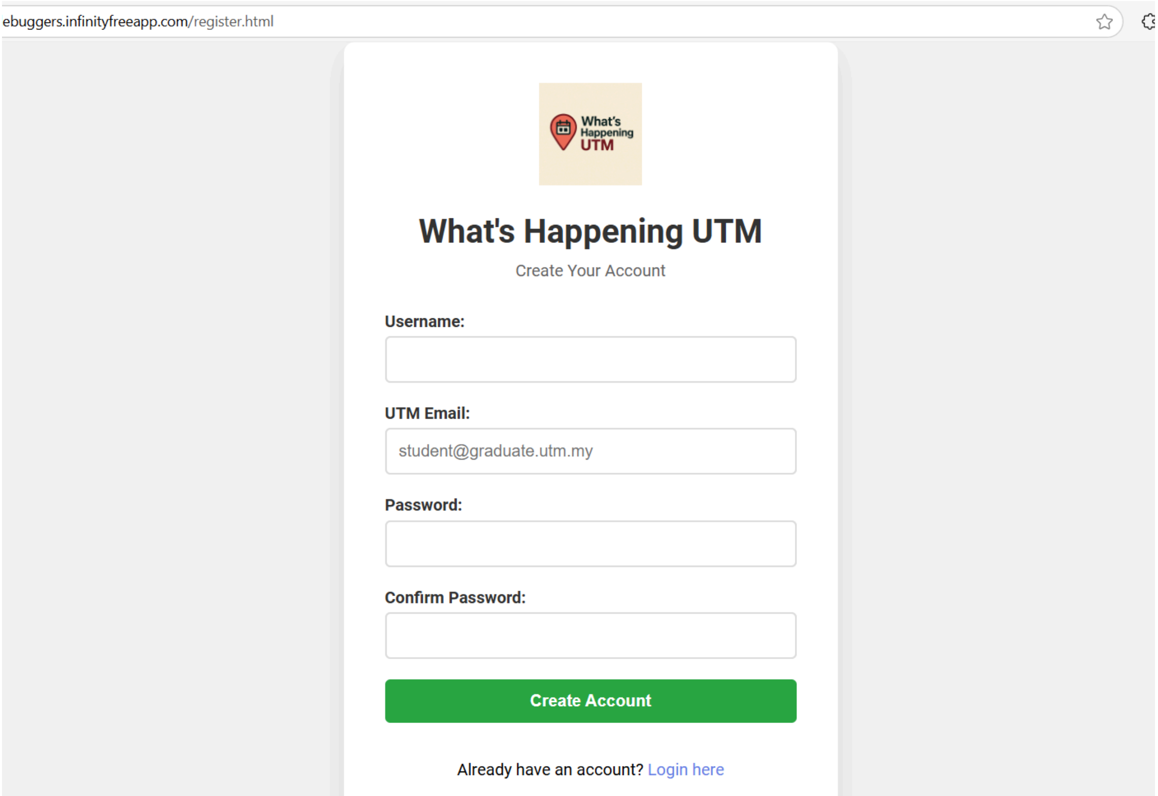
Step 5 : Accessing the Live Website

Once uploaded and configured , your project is live at the provided URL such as : https://group4debuggers.infinityfreeapp.com

8.0 Screenshots

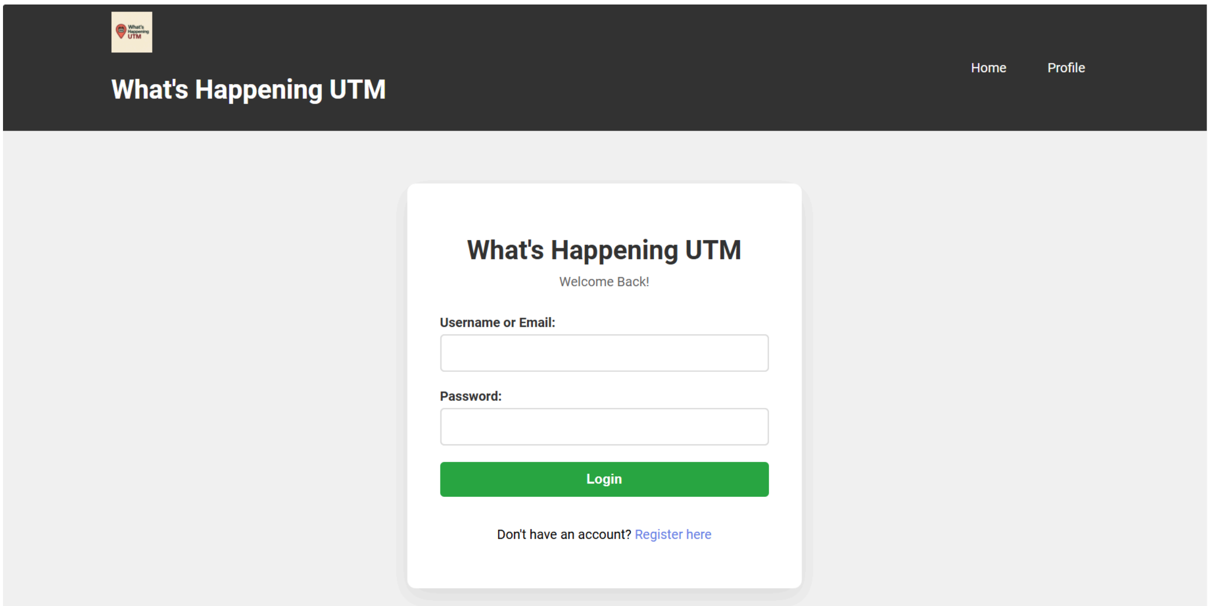
Registration page

This page allows new users to create an account by entering their details. Input validation ensures all required fields are filled before submission.



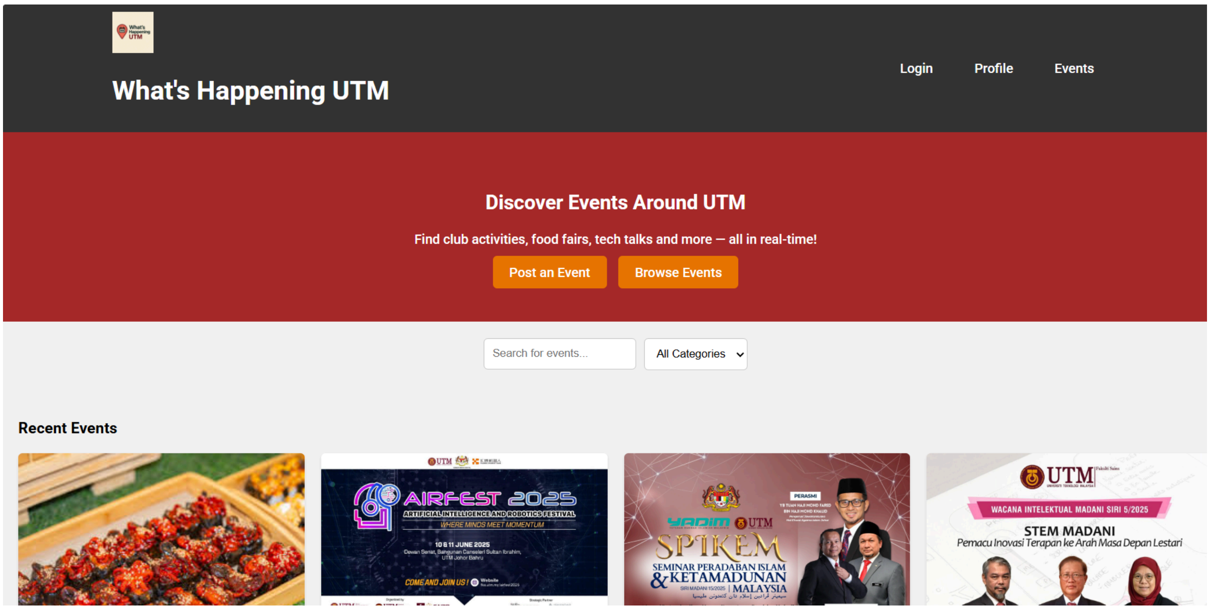
Login page

Registered users can log in with their email and password. The system includes error handling for incorrect credentials and missing input.



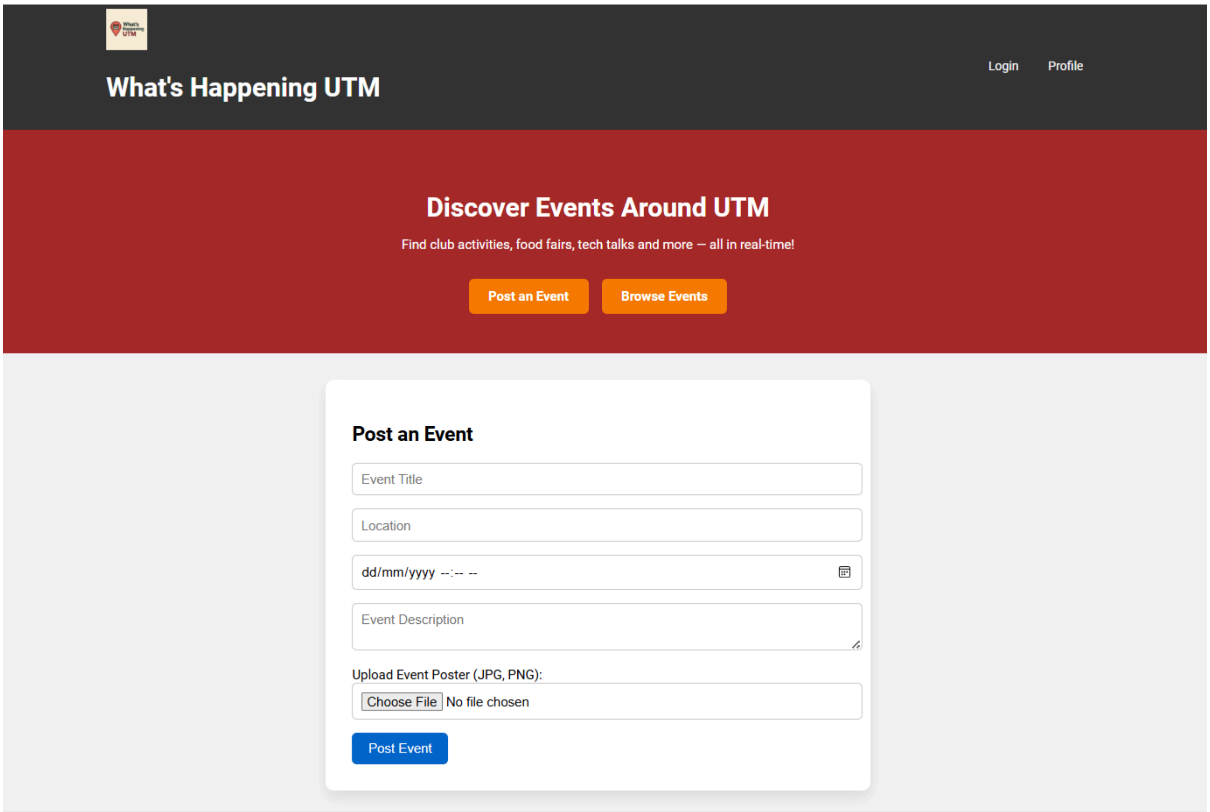
Homepage

The homepage displays a dynamic list of upcoming events posted by students.



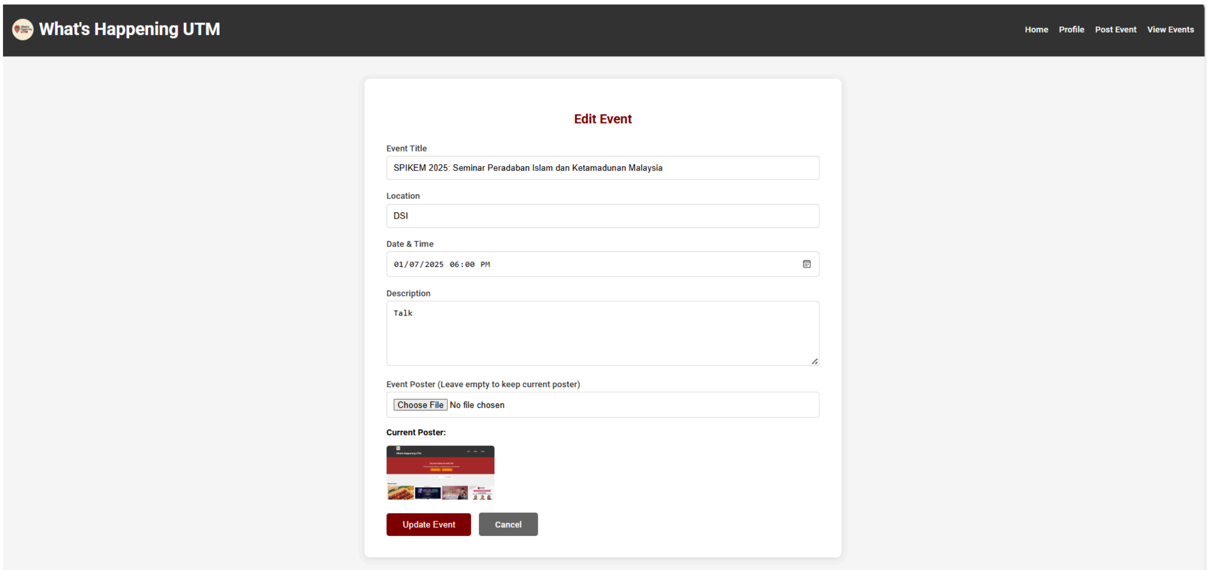
Create Event page

Users can submit a new event with a title, description, date, time, category, location, and an image upload. This form includes validation and is accessible only to logged-in users.



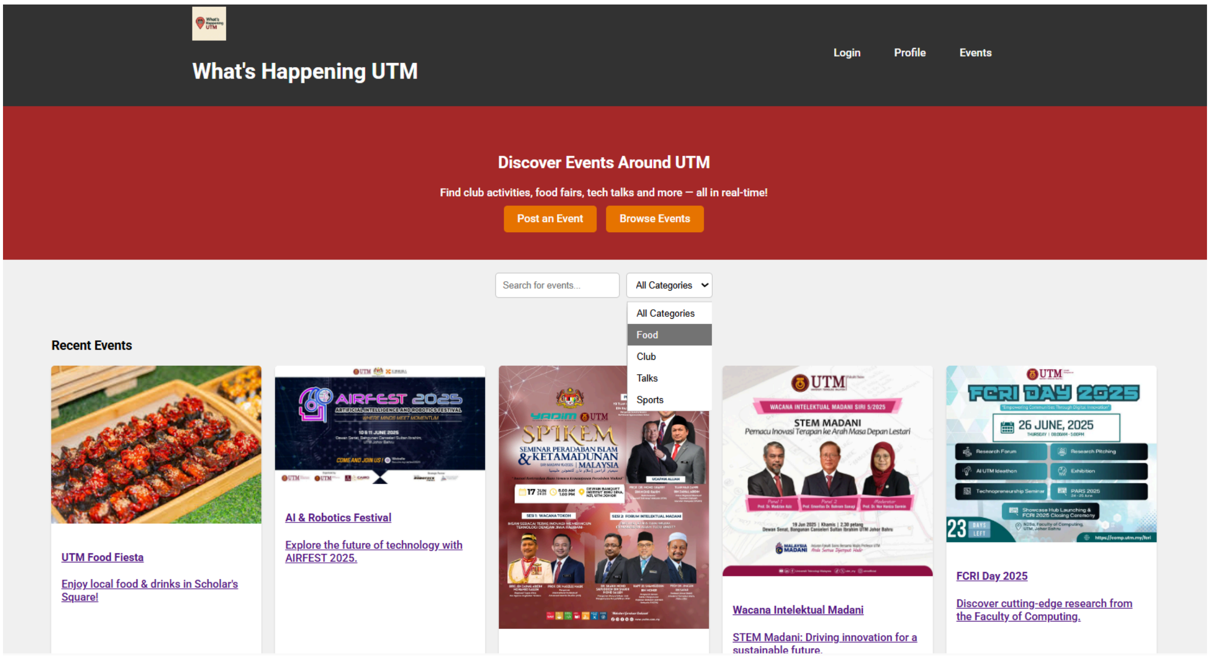
Edit page

Students are allowed to make changes and update the events.



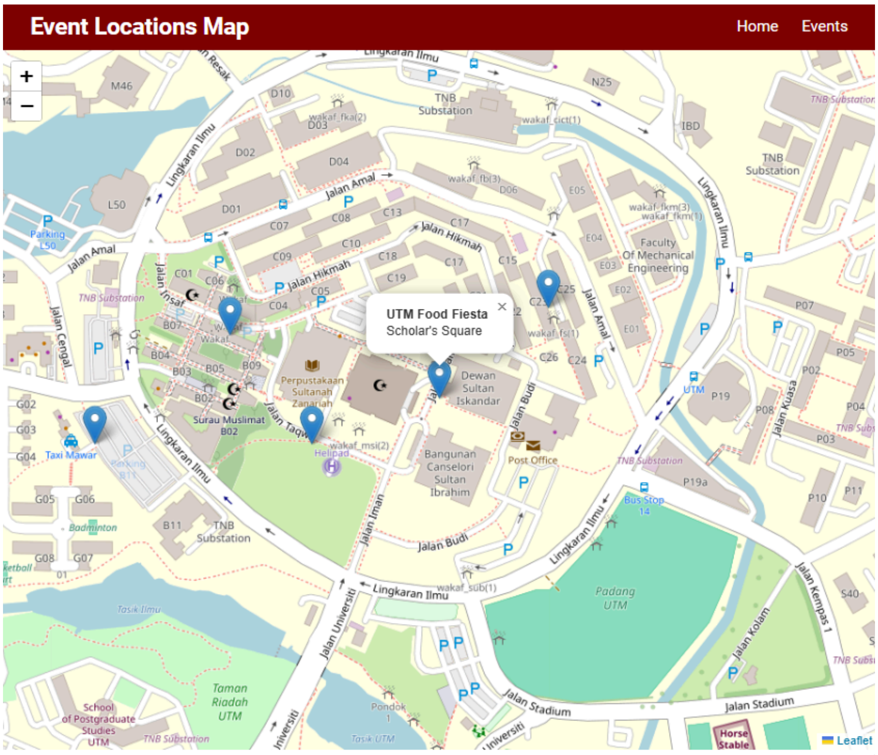
Event Filter/Search page

Students can refine event results using filters by category (e.g., food fair, academic talk) or by search terms. This enhances the event discovery experience.



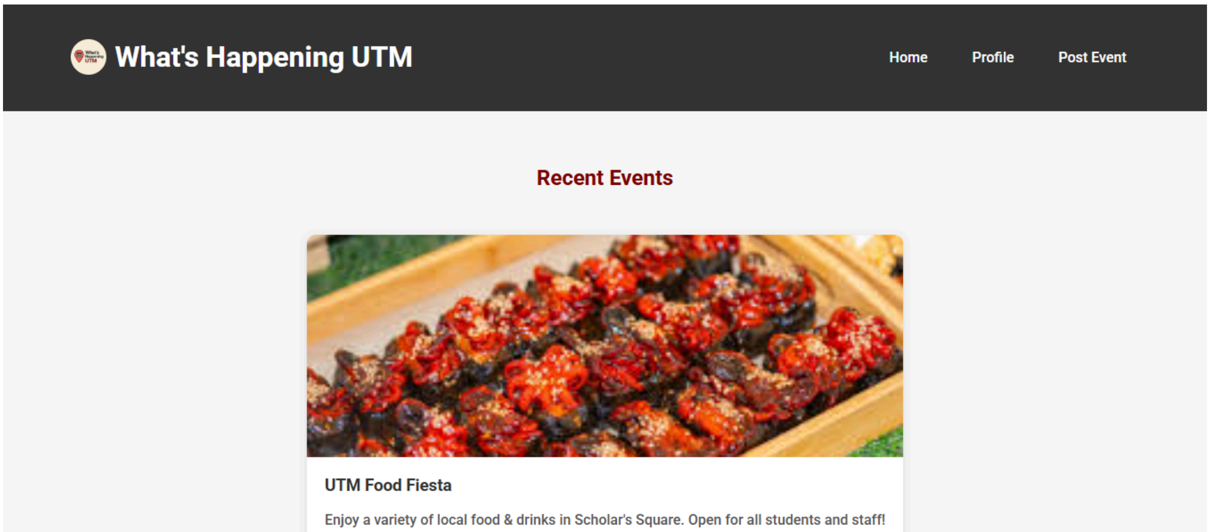
Google Maps view

Events are pinned to an interactive Google Map, allowing users to view locations visually. Clicking on a pin reveals event details.



Events page

In this page, it will show the recent events and also events that have been posted. It also have the description about the event.



9.0 Conclusion

The development and deployment of “What’s Happening UTM ?” represent a greater accomplishment than simply completing the requirements for a Web Programming course. It represents the ability of a focused, small team to take a particular problem within the campus setting and turn it into a complete, production-quality solution. Throughout the project, the team progressed through problem definition, design, programming, user testing, and documentation, while refining both their technical abilities and cooperative practices in the process. Functionally, the application has achieved its primary aims. Student users reported that finding events is quicker and more intuitive than browsing through social media whereas organizers appreciated the convenience of being able to change details or correct errors without the need to reprint posters or re-post messages. Load testing with example data proved that page rendering and search functions remained effective even with hundreds of concurrent records, thus validating the effectiveness of utilizing lightweight PHP along with properly indexed MySQL tables. From a learning standpoint, the group gained a more nuanced appreciation for web security (password hashing, preventing session hijacking), principles of responsive design, and integrating with APIs. Problems faced from cross-browser CSS incompatibilities to race conditions with concurrent post edits offered fertile ground to hone debugging skills and version-control best practices. These proficiencies will transfer to future software projects, be they academic research or industry internship. The future advances include a progressive web app layer for offline caching, QR code reading for rapid event check-ins, machine learning recommendations to prioritize personalized recommendations, and interaction with campus businesses to enhance campus life. "What's Happening UTM?" exemplifies the power of user-centered design in addressing common challenges and bringing communities together. By combining technical knowledge with a desire to improve student experience, the project achieves its ultimate goal of changing information into engagement and engagement into a more vibrant UTM.

10.0 Acknowledgement

To complete this project, we received help and support from many individuals and resources along the way. We would like to take this opportunity to express our appreciation to all who contributed, directly or indirectly, to the success of our project titled “What’s Happening UTM?”. First and foremost, we would like to thank our lecturer, Ts. Dr. Sarina binti Sulaiman, for her continuous guidance, constructive feedback, and encouragement throughout the course. Her expertise in web programming and her clear explanations helped us better understand full-stack development and apply it effectively in building our event bulletin web application. We also want to extend our sincere appreciation to all our group members — Nuur Aisyah binti Ruzi, Nurul Natalia binti Rosnizam, Adlina Narisya binti Ismail, and Koo Xuan — for their strong teamwork, cooperation, and dedication. Each member played a crucial role, from frontend and backend development to database design, UI improvements, and report writing. The successful completion of this project is the result of everyone’s collective effort. In addition, we are grateful for the online resources and tutorials that enhanced our understanding of various web development technologies, including PHP, MySQL, JavaScript, and the integration of APIs. References such as W3Schools and developer documentation provided valuable support during the development and debugging phases. Overall, this project has been a meaningful and enriching experience. We are thankful for the opportunity to apply classroom knowledge in a practical setting and to improve both our technical and collaborative skills. Thank you to everyone who supported us throughout this journey.

11.0 References

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